| Reg No. | 4 | 4 | 1 | 3 | 0 | 1 | 7 | 2 |
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SATHYABAMA
INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

CONTINUOUS ASSESSMENT EXAM-1

Programme: B.E-ECE
Course Code: SECB1301
Course Name: Electronic Circuits
Batch: 2024-2028

Max. Marks: 50
Time: 2 Hours
Sem: III
Date: 5.8.25

Part-A Answer ALL the questions (7 × 2=14)

| Q.No | Questions | CO |
|---|---|---|
| 1. | Sketch the circuit diagram for bridge rectifier. | 1 |
| 2. | Specify the types of rectifier. | 1 |
| 3. | Determine the d.c power delivered to the load if the d.c current is 20.7mA and load resistance is 900Ω. | 1 |
| 4. | Sketch the capacitor filter circuit. | 1 |
| 5. | Mention the different methods of biasing. | 2 |
| 6. | Delineate the stability factor. | 2 |
| 7. | Specify the need for biasing. | 2 |

Part-B Answer ALL the questions (3 × 12 = 36)

| | | |
|---|---|---|
| 8. | Examine the operation of half wave rectifier with relevant waveforms, ripple factor, efficiency and transformer utilization factor. | 1 |
| | (OR) | |
| 9. | Analyze the ripple factor for a inductor filter using full wave rectifier. | 1 |
| 10. | Interpret the ripple factor, efficiency and peak factor of full wave rectifier with circuit diagram and relevant waveforms. | 1 |
| | (OR) | |
| 11. | Analyze the ripple factor for a CLC filter using full wave rectifier. | 1 |
| 12. | Interpret the base current and collector current for fixed bias circuit and voltage divider bias circuit. | 2 |
| | (OR) | |
| 13. | Examine the stability factor for fixed bias circuit and collector to base bias circuit. | 2 |